

Math 208 - Tutorial 1

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Introduction

LaTeX's features for typesetting mathematics make it a compelling choice for writing technical documents. This article shows the most basic commands needed to get started with writing maths using LaTeX.

Writing basic equations in LaTeX is straightforward, for example:

```
1 \documentclass{article}
2 \begin{document}
3
4 The well-known Pythagorean theorem  $x^2 + y^2 = z^2$  was
5 proved to be invalid for other exponents.
6 Meaning the next equation has no integer solutions:
7
8  $[x^n + y^n = z^n]$ 
9
10 \end{document}
```

The well-known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents. Meaning the next equation has no integer solutions:

$$x^n + y^n = z^n$$

As you see, the way the equations are displayed depends on the delimiter, in this case $[\dots]$ and $(\dots)$.

Mathematical Modes

LaTeX allows two writing modes for mathematical expressions: the inline math mode and the display math mode:

- **Inline math mode** is used to write formulas that are part of a paragraph.
- **Display math mode** is used to write expressions that are not part of a paragraph, and are therefore put on separate lines.

Inline Math Mode

You can use any of these "delimiters" to typeset your math in inline mode:

- ...
- $\dots$
- $\begin{math}...\end{math}$

They all work, and the choice is a matter of taste. Here are some examples:

```

1  \documentclass{article}
2  \begin{document}
3
4  \noindent Standard \LaTeX{} practice is to write inline math by enclosing it
   ↪  between \verb|\(...\)|:
5
6  \begin{quote}
7  In physics, the mass-energy equivalence is given
8  by the equation \((E=mc^2)\), discovered in 1905 by Albert Einstein.
9  \end{quote}
10
11 \noindent Instead of writing inline math between \verb|\(...\)|, you can use
   ↪  \texttt{\$...\$}:
12
13 \begin{quote}
14 In physics, the mass-energy equivalence is given
15 by the equation  $E=mc^2$ , discovered in 1905 by Albert Einstein.
16 \end{quote}
17
18 \noindent Or, you can use \verb|\begin{math}...\end{math}|:
19
20 \begin{quote}
21 In physics, the mass-energy equivalence is given
22 by the equation \begin{math}E=mc^2\end{math}, discovered in 1905 by Albert
   ↪  Einstein.
23 \end{quote}

```

24
25 `\end{document}`

Standard L^AT_EX practice is to write inline math by enclosing it between `\(...\)`:

In physics, the mass-energy equivalence is stated by the equation $E = mc^2$, discovered in 1905 by Albert Einstein.

Instead of writing inline math between `\(...\)`, you can use `$...$`:

In physics, the mass-energy equivalence is stated by the equation $E = mc^2$, discovered in 1905 by Albert Einstein.

Or, you can use `\begin{math}...\end{math}`:

In physics, the mass-energy equivalence is stated by the equation $E = mc^2$, discovered in 1905 by Albert Einstein.

Display Math Mode

Use one of these constructions to typeset math in display mode:

- `\[...\]`
- `\begin{displaymath}...\end{displaymath}`
- `\begin{equation}...\end{equation}`

Display math mode has two versions, which produce numbered or unnumbered equations. Here's a basic example:

```
1 \documentclass{article}
2 \begin{document}
3
4 The mass-energy equivalence is described by the famous equation
5
6 \[E=mc^2\]
7
8 discovered in 1905 by Albert Einstein.
9 In natural units ( $c = 1$ ), the formula expresses the identity
10
11 \begin{equation}
12 E = m
13 \end{equation}
14
15 \end{document}
```

Another Example

```
1 \documentclass{article}
2 \usepackage{amsmath}
3 \begin{document}
4
5 This is a simple math expression  $\sqrt{x^2+1}$  inside text.
6 And this is also the same:
7 \begin{math}
8 \sqrt{x^2+1}
9 \end{math}
10 but by using another command.
11
12 This is a simple math expression without numbering:
13 
$$\sqrt{x^2+1}$$

14
15 This is also the same:
16 \begin{displaymath}
17 \sqrt{x^2+1}
18 \end{displaymath}
19
20 \ldots and this:
21 \begin{equation*}
22 \sqrt{x^2+1}
23 \end{equation*}
24
25 \end{document}
```

Reference Guide

Below is a table with some common math symbols. For a more complete list, see the List of Greek letters and math symbols.

Description	Code	Examples
Greek letters	<code>\alpha \beta \gamma \rho</code>	$\alpha, \beta, \gamma, \rho$
Binary operators	<code>\times \otimes \oplus</code>	$\times, \otimes, \oplus$
Relation operators	<code>< > \subset \supset</code>	$<, >, \subset, \supset$
Others	<code>\int \sum \prod</code>	$\int, \sum, \prod$

Table

Simple Table

```
1 \begin{table}[h!]  
2 \centering  
3 \begin{tabular}{|c|c|c|}  
4 \hline  
5 Header 1 & Header 2 & Header 3 \\ \hline  
6 Cell 1 & Cell 2 & Cell 3 \\ \hline  
7 Cell 4 & Cell 5 & Cell 6 \\ \hline  
8 \end{tabular}  
9 \caption{A simple table with three columns.}  
10 \end{table}
```

Header 1	Header 2	Header 3
Cell 1	Cell 2	Cell 3
Cell 4	Cell 5	Cell 6

Table 1: A simple table with three columns.

Table with Different Alignments

```
1 \begin{table}[h!]  
2 \centering  
3 \begin{tabular}{|l|r|c|}  
4 \hline  
5 Left Aligned & Right Aligned & Centered \\ \hline  
6 Item 1 & 123 & Yes \\ \hline  
7 Item 2 & 456 & No \\ \hline  
8 \end{tabular}  
9 \caption{A table with different column alignments.}  
10 \end{table}
```

Left Aligned	Right Aligned	Centered
Item 1	123	Yes
Item 2	456	No

Table 2: A table with different column alignments.

Table with Multi-row and Multi-column

```

1 \begin{table}[h!]
2 \centering
3 \begin{tabular}{|c|c|c|}
4 \hline
5 \multirow{2}{*}{Merged Row} & Column 2 & Column 3 \\
6 & Column 2a & Column 3a \\
7 Column 1 & \multicolumn{2}{|c|}{Merged Columns} \\
8 \end{tabular}
9 \caption{A table with merged rows and columns.}
10 \end{table}

```

Merged Row	Column 2	Column 3
	Column 2a	Column 3a
Column 1	Merged Columns	

Table 3: A table with merged rows and columns.

Table with Adjustable Column Width

```

1 \begin{table}[h!]
2 \centering
3 \begin{tabularx}{\textwidth}{|X|X|X|}
4 \hline
5 Header 1 & Header 2 & Header 3 \\
6 This is a long cell text that will wrap automatically & \\
7 Another long text & \\
8 Short text \\
9 \end{tabularx}
10 \caption{A table with automatically adjusted column width.}
11 \end{table}

```

Header 1	Header 2	Header 3
This is a long cell text that will wrap automatically	Another long text	Short text

Table 4: A table with automatically adjusted column width.

Table with Colored Rows

```
1 \begin{table}[h!]  
2 \centering  
3 \begin{tabular}{|c|c|c|}  
4 \hline  
5 \rowcolor[HTML]{EFEFEF}  
6 Header 1 & Header 2 & Header 3 \\ \hline  
7 Cell 1 & Cell 2 & Cell 3 \\ \hline  
8 \rowcolor[HTML]{CFE2F3}  
9 Cell 4 & Cell 5 & Cell 6 \\ \hline  
10 \end{tabular}  
11 \caption{A table with alternating row colors.}  
12 \end{table}
```

Header 1	Header 2	Header 3
Cell 1	Cell 2	Cell 3
Cell 4	Cell 5	Cell 6

Table 5: A table with alternating row colors.

Examples

Math Mode Examples

All math commands must be typed in math mode. This equation is not typed in math mode:
 $a + b = c$.

This equation is typed in math mode: $a + b = c$.

Inline Math Mode Inline math mode allows us to insert equations into our paragraphs, such as:

$$\int_1^5 \sin(x) \cos(x) dx, (\int_1^5 \sin(x) \cos(x) \, dx)$$

Display Math Mode Display math mode is used for formulas not embedded in paragraphs. The equations will be put on a separate line:

$$\frac{d}{dx} x^2 = 2x, (\frac{d}{dx} x^2 = 2x)$$

$$\int \cos(x) dx = \sin(x) + C, (\int \cos(x) \, dx = \sin(x) + C)$$

$$\sum_{i=1}^{\infty} \sqrt{i} = \sqrt{1} + \sqrt{2} + \sqrt{3} + \dots$$

$$\sum_{i=1}^{\infty} \sqrt{i} = \sqrt{1} + \sqrt{2} + \sqrt{3} + \dots$$

$$f(x) = \beta_0 + \beta_1 x, \quad \mathbf{A}\vec{x} = \vec{b}$$

$$f(x) = \beta_0 + \beta_1 x, \quad \mathbf{A}\vec{x} = \vec{b}$$

Matrix Data

Here is an example of matrix data:

```

1  \[
2  \begin{matrix}
3  1 & 2 & 3 & 4 & 5 & 6 \\
4  1 & 2 & 3 & 4 & 5 & 6 \\
5  1 & 2 & 3 & 4 & 5 & 6 \\
6  \end{matrix}
7  \]
8
9  A more structured matrix:
10 \[
11 \begin{bmatrix}
12 1 & 2 & 3 \\
13 4 & 5 & 6 \\
14 7 & 8 & 9 \\
15 \end{bmatrix}
16
17 Or:
18
19 \begin{pmatrix}
20 1 & 2 & 3 \\
21 \end{pmatrix}
22 =
23 \begin{bmatrix}
24 15 & 32 \\
25 \end{bmatrix}
26 \]
27
```

1	2	3	4	5	6
1	2	3	4	5	6
1	2	3	4	5	6

A more structured matrix:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Or:

$$(1 \ 2 \ 3) = \begin{bmatrix} 15 \\ 32 \end{bmatrix}$$

Constraint Optimization Notation

Let x be the x -distance from the origin and y be the y -distance from the origin. The decision variables are the (x, y) coordinates, denoted by x and y respectively.

```

1  Minimize:
2  \[
3  x + y
4  \]
5
6  Subject to:
7  \[
8  y + 5x = 800
9  \]
10 \[
11 x - xy + 40y = 0
12 \]
13
```

Minimize:

$$x + y$$

Subject to:

$$y + 5x = 800$$

$$x - xy + 40y = 0$$